

Jamii Savings Integration Platform — Solution Design Document

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1. Document control and design summary

Field	Value
Version	1.4 (Final draft)
Author	Emmanuel
Implements	BRD v1.4, Jamii Savings Integration Platform
Status	Final draft

Three MI integrations sit behind one APIM gateway, and one Jenkins pipeline deploys all of them to a docker-compose stack. Every API returns the same error contract and masks sensitive data in its logs.

Key decisions

Area	Decision
Balance backend	H2 2.x in server (TCP) mode in its own container, exposed through an MI Data Service with parameterised queries
Customer backend	Jamii mock backends (Spring Boot 3.5, JDK 21): GET /customers/{id} with bank-shaped records and rule-driven 404, 500 and delay; in mock-backends/ of the same repository
Loan backend	Jamii mock backends: credit-check SOAP service (CheckEligibility) with rule-driven client faults, server faults and delay
Security	OAuth2 for Balance and Loan; API Key for Customer
Throttling	Balance and Customer on Gold; Loan on a custom LoanCheck10PerMin tier
Gateway policy	Custom APIM operation policy on the Balance API: correlation ID and channel header injection
Packaging	Every component ships as a versioned Docker image (MI with CARs, APIM, H2, ELK, Jenkins agent), pushed to a local registry; the mock backends are built and run in the same stack
Release safety	Validate everything first, deploy, smoke test, roll back on any failure
Edge	nginx as the single public entry, TLS with a wildcard certificate, one subdomain per service under jamii.daqiqtech.co.ke (section 8.3); stretch goal

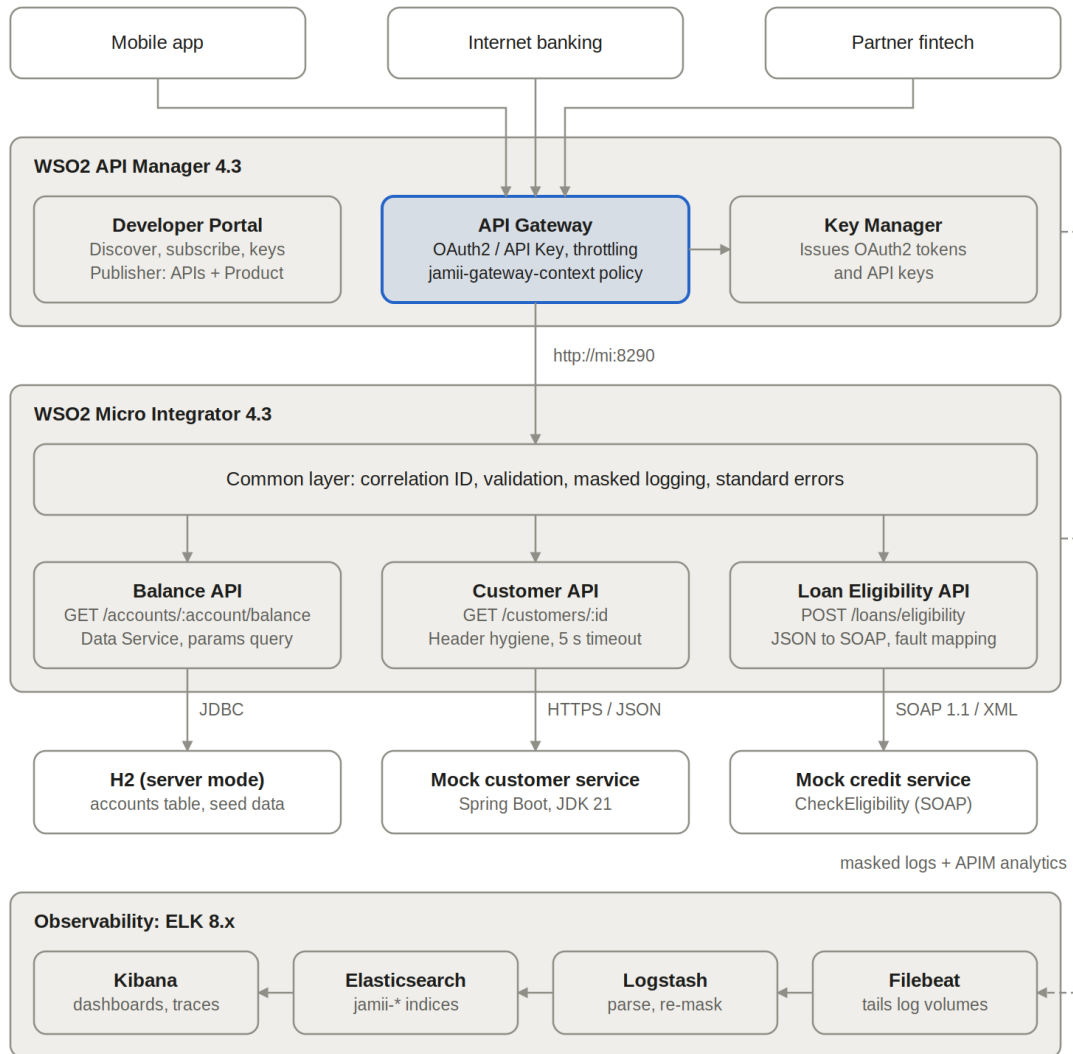
Observability: MI and APIM logs flow through Filebeat and Logstash into Elasticsearch. Kibana provides dashboards and end-to-end tracing by correlation ID (section 8.2). ELK and the nginx edge are stretch goals, started only after the core pipeline is green, to hold the 12 Oct submission.

Deployment model: Option C, self-managed. APIM and MI both run as self-managed containers, not on WSO2's managed cloud. Their images are built with docker compose build from product distributions downloaded from wso2.com, using our own Dockerfiles. The team therefore owns hosting, patching, keystores, credentials and backups (section 8.1).

2. Architecture overview

Consumers only ever talk to the APIM gateway, through nginx at api.jamii.daqiqtech.co.ke. MI is an internal layer that normalises three very different backends behind one contract.

Every consumer call passes nginx and the APIM gateway before it reaches MI



Logical architecture: consumers, APIM, MI, backends and ELK

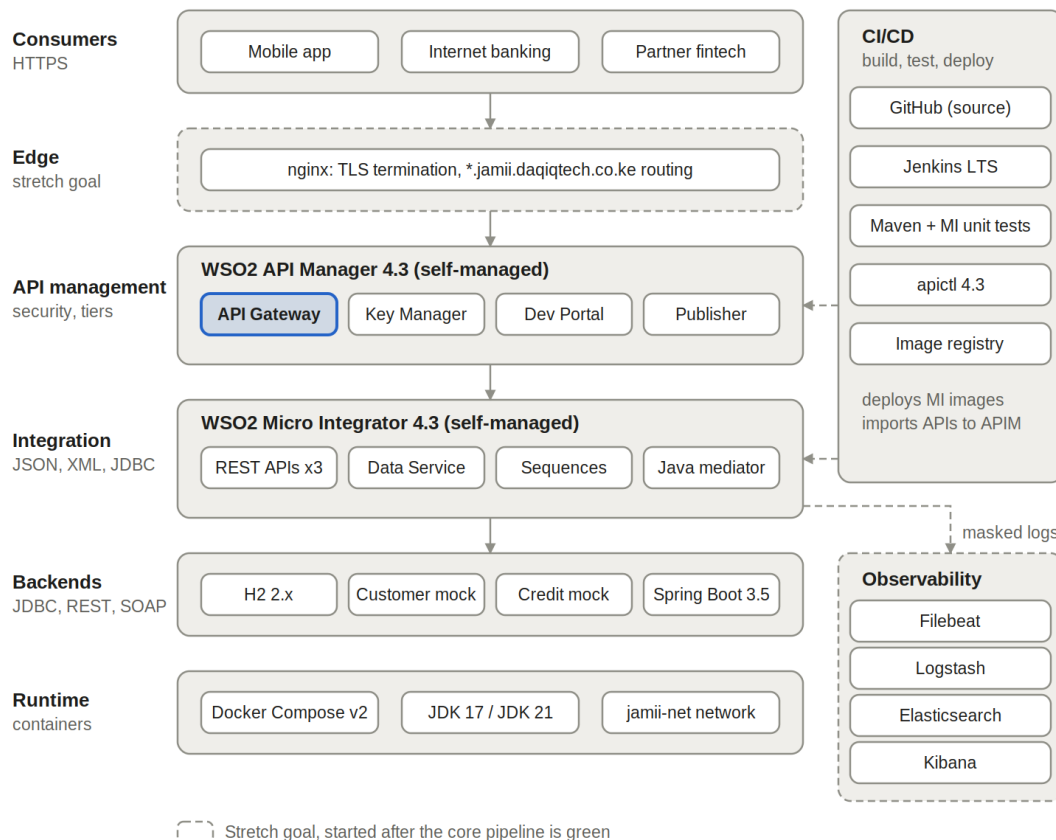
The gateway handles identity, quotas and edge policy. MI handles validation, transformation, masking and error mapping. The backends never see consumer credentials. APIM and MI both ship masked logs to the ELK stack (dashed line).

2.1 High-level design and technologies

The HLD shows the solution as layers, each with the technology that implements it. Requests flow down through the layers. CI/CD deploys into the two WSO2 layers from the side, and observability collects masked logs from the side.

One gateway, one integration layer, one pipeline

Jamii Savings high-level design, with the technology used at each layer



High-level design: layers and technologies

3. Technology stack

These are the target versions. On Day 1, confirm the latest matching MI/APIM/apictl release and pin it in .env.

Layer	Tool	Version (pin)	Purpose
Integration	WSO2 Micro Integrator	4.3.0 (on JDK 17)	Mediation, data services
Tooling	WSO2 MI VS Code extension + Maven 3.9	latest / 3.9.x	Project, CAR build, unit tests
API management	WSO2 API Manager (all-in-one)	4.3.0 (on JDK 17)	Publisher, Dev Portal, Gateway, Key Manager
CLI	apictl	4.3.x	Export/import APIs and products per environment
Database	H2	2.x, server (TCP) mode	Accounts data
Log shipping	Filebeat	8.x	Tails MI and APIM log volumes
Log processing	Logstash	8.x	Parse, enrich, re-mask, route to indices
Search and storage	Elasticsearch	8.x, single node	Log and APIM analytics indices
Dashboards	Kibana	8.x	Operations dashboard, trace by correlation ID
Containers	Docker, Compose v2, registry:2	latest	Build, tag, store and run every image
CI/CD	Jenkins LTS (custom image: JDK 21 and JDK 17, Maven, apictl, curl, jq, Docker CLI)	LTS	Declarative pipeline
Tests	MI unit testing framework,	latest	Functional, contract, schema

Layer	Tool	Version (pin)	Purpose
	gateway-check.sh (curl)		tests
Edge	nginx	stable (pinned)	TLS termination, host-based routing, IP allowlists
Backends	Jamii mock backends (Spring Boot, Spring Web Services)	3.5.x, JDK 21, Maven, jar	Customer REST and credit-check SOAP mocks with rule-driven faults and delays
Java runtimes	Eclipse Temurin	17 (WSO2 MI, APIM); 21 (H2, mock backends, Jenkins)	The WSO2 start scripts accept JDK 11 to 17 only

All four Elastic components are pinned to the same minor version.

4. Component design per API

Each API is an MI REST API artifact that calls shared sequences for validation, logging and errors. Only the backend call and the transformation differ.

4.1 Balance API (/accounts/{accountNumber}/balance)

Artifact	Name	Responsibility
Datasource	JamiiAccountsDS	JDBC pool to H2 at jdbc:h2:tcp://h2:9092/jamii (H2 driver jar baked into the MI image); credentials from env vars
Data Service	AccountsDataService	Query getAccountByNumber: SELECT account_number, status, balance, currency FROM accounts WHERE account_number = ?
REST API	BalanceAPI	Validates path, calls data service, builds envelope
Fault sequence	Common_FaultSeq	Maps SQL/connection errors to 503 DB_UNAVAILABLE

Flow: validate `^[0-9]{10,16}$` (else 400), call the data service. Zero rows returns 404 ACCOUNT_NOT_FOUND. One row is transformed with a PayloadFactory into the BRD envelope, with `requestId` = correlation ID and `timestamp` = UTC now. A JDBC or connection error triggers the fault sequence and returns 503.

4.2 Customer API (/customers/{customerId})

Artifact	Name	Responsibility
HTTP endpoint	CustomerBackendEP	http://jamii-mock-backends:8080/customers/{uri.var.customerId} (base URL from CUSTOMER_BACKEND_URL); timeout 5000 ms, action fault
REST API	CustomerAPI	Header manipulation, pass-through body, 404 mapping
Sequence	Common_HeaderHygiene	Add/propagate X-Correlation-ID; strip X-Internal-*, Server, X-Powered-By, Set-Cookie

Flow: validate `^[0-9]{1,10}$` (else 400), apply header hygiene and call the endpoint. A 200 body is passed through and the X-Correlation-ID response header is set. A backend 404 returns 404 CUSTOMER_NOT_FOUND. A timeout returns 504 BACKEND_TIMEOUT. Connection refused or DNS failure triggers the fault sequence and returns 503 BACKEND_UNAVAILABLE. The backend URL is a config property, so it can point at any deployed instance of the mock backends without a code change.

Mock customer rules. The Jamii mock backends service returns records shaped like a Kenyan bank's customer file, so masking is exercised on national ID and phone, not just names. Every response also carries Server, X-Powered-By and X-Internal-* headers, which MI must strip.

Request	Mock response	Purpose
GET /customers/1001 to 1005	200 with customerId, fullName, nationalId, phone (+254), email, kycStatus, branch	Happy path and PII masking
GET /customers/9999 (or any unknown ID)	404	CUSTOMER_NOT_FOUND mapping
GET /customers/5000	500	502 BAD_GATEWAY mapping
GET /customers/5004	200 after a 7 s delay	Exceeds the 5 s timeout, so 504 BACKEND_TIMEOUT

The mock backends are a separate Spring Boot project with its own Dockerfile and docker-compose file, built and deployed independently of the integration stack (section 8.4).

4.3 Loan Eligibility API (POST /loans/eligibility)

Request:

```
{ "customerId": "7", "monthlyIncome": 120000, "existingMonthlyDebt": 15000,
  "requestedMonthlyInstalment": 25000 }
```

Mapping to the credit-check SOAP operation CheckEligibility (namespace <http://jamii.daqiqtech.co.ke/credit/v1>, WSDL at /ws/creditCheck.wsdl):

- The four JSON fields map one-to-one to the SOAP request elements, built with a PayloadFactory.
- The credit engine computes debt-to-income % = (existing debt + instalment) × 100 ÷ income, and approves at 40% or less.
- MI does not make the decision. It mediates the protocol, maps decision, debtToIncomePercent, maxAllowedPercent and referenceId back to JSON, derives eligible, and maps faults.

Response:

```
{ "customerId": "7", "decision": "ELIGIBLE", "eligible": true, "debtToIncomePercent": 33,
  "maxAllowedPercent": 40, "creditReference": "CRD-7F3A9C21", "requestId": "...", "timestamp":
  "..." }
```

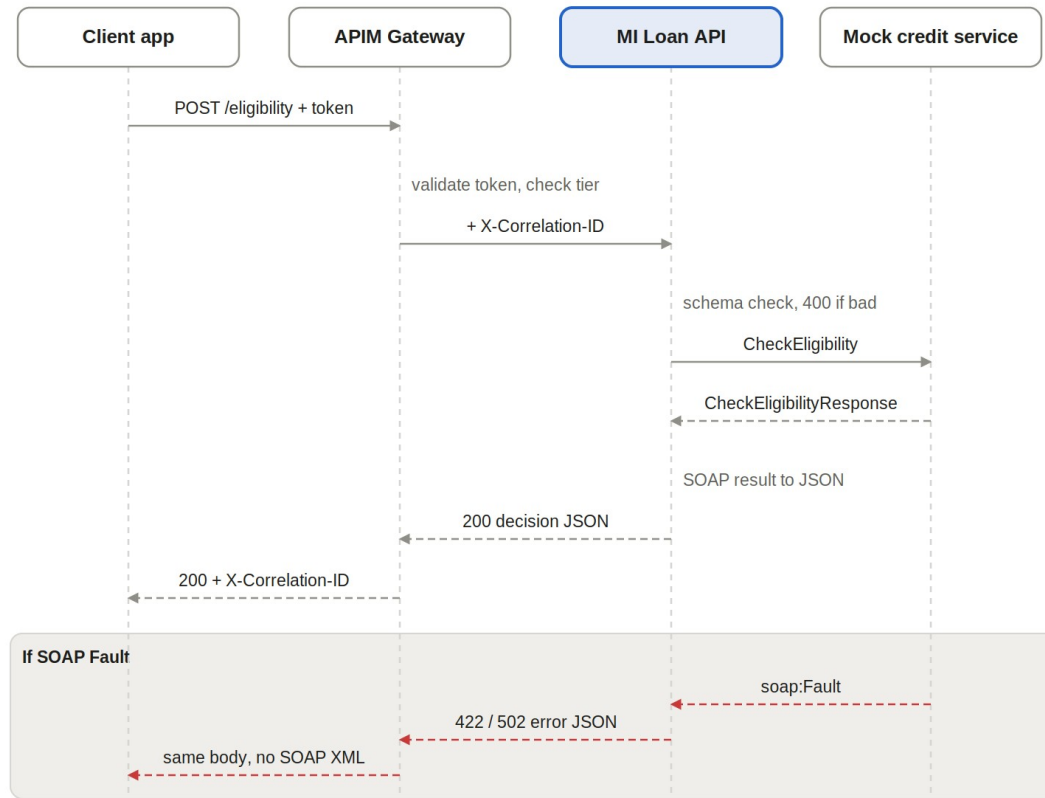
Why a purpose-built SOAP mock: a credit-check operation gives the SOAP call real lending meaning, and the mock produces both kinds of SOAP fault, plus a slow answer, on demand. The trade-off is that it is a service we built rather than an existing public one, so the README states this and justifies it.

SOAP outcome	Trigger	REST response
CheckEligibilityResponse	Normal values	200 with decision
soap:Client INVALID_INCOME	monthlyIncome 0	422 ELIGIBILITY_REJECTED, "Monthly income must be greater than zero"
soap:Client CUSTOMER_BLACKLISTED	customerId 6000	422 ELIGIBILITY_REJECTED, "Applicant is not eligible for credit"
soap:Server	customerId 5000	502 BAD_GATEWAY (fault string logged, never returned)
No answer within 10 s	customerId 5004 (12 s delay)	504 BACKEND_TIMEOUT
Unreachable	Mock backends down	503 BACKEND_UNAVAILABLE

5. Loan Eligibility request flow

This is the most complex flow, because it crosses protocols twice: JSON to SOAP, then SOAP back to JSON.

A SOAP fault never reaches the client as XML



Loan Eligibility sequence: happy path and SOAP fault path

Solid arrows are requests and dashed arrows are responses. Red marks the fault path, where MI's fault sequence turns the SOAP fault into the standard error JSON before anything is returned.

6. Cross-cutting design

Shared logic lives in four reusable sequences and one template, so each API stays thin and every API behaves the same way.

Artifact	Type	What it does
Common_InSeq	Sequence	Reads or generates X-Correlation-ID (UUID v4), stores it as property correlationId, records start time
Common_MaskedLog	Sequence template	Logs a stage (REQUEST_IN, BACKEND_CALL, RESPONSE_OUT) with masked fields; params: stage, api, extra
Common_ErrorResponse	Sequence template	Builds the standard error JSON from params (httpStatus, code, message, details), sets status and correlation header, responds
Common_FaultSeq	Fault sequence	Catches unhandled errors, classifies them by ERROR_CODE (timeout 101504, connection 101503, SQL) and calls Common_ErrorResponse
Common_HeaderHygiene	Sequence	Strips internal and server-identifying headers inbound and outbound

Validation: each API validates input before any backend call. Path parameters use regex. The loan body uses a JSON Schema (loan-request.schema.json, stored as a registry resource) via the Validate mediator. A failure returns 400 VALIDATION_ERROR with field-level details. Malformed JSON is caught in the fault sequence and also returns 400, never 500.

Masking rules: masking happens in two layers, with no Java code. First, Common_MaskedLog only logs whitelisted values that the API has already masked (the account number reduced to its last 4 digits, the customer ID, status codes); bodies and amounts are never logged. Second, every MI log pattern passes the message through a log4j2 %replace rule that masks any run of 10 or more digits to ***** plus the last 4. The second layer also covers MI's own components, such as data-service fault messages and transport errors, which log request URLs and parameters.

Field	Rule	Example
accountNumber	keep last 4	*****7890
name, username, email, phone, address	full mask	***
monthlyIncome, debts, instalment	omitted from logs	n/a

Logging format: one line per stage, key=value, for example:

```
stage=REQUEST_IN api=BalanceAPI correlationId=5b1f... account=*****7890 method=GET
stage=BACKEND_CALL api=BalanceAPI correlationId=5b1f... backend=AccountsDataService
stage=RESPONSE_OUT api=BalanceAPI correlationId=5b1f... httpStatus=200 latencyMs=42
```

7. API Manager design

All three APIs are published from apictl projects in Git, never hand-edited in the Publisher. Git is the source of truth.

API	Context / version	Security	Subscription tier	Backend (MI)
Jamii Account Balance	/jamii/accounts/v1	OAuth2	Gold	http://mi:8290/accounts
Jamii Customer Profile	/jamii/customers/v1	API Key	Gold	http://mi:8290/customers
Jamii Loan Eligibility	/jamii/loans/v1	OAuth2	LoanCheck10PerMin (custom; tier names may not contain hyphens)	http://mi:8290/loans

Tier reason (for README): Loan eligibility gets a tighter custom tier because each call fans out to a slow, external SOAP dependency with no SLA, and the tier protects it from overload.

Security reason: OAuth2 suits user-context and financial data (Balance, Loan). API Key shows the simpler server-to-server option for partner lookups (Customer). Using both shows the trade-off.

Custom mediation policy: jamii-gateway-context is an APIM 4.x operation policy (.j2 + .yaml) applied on the request flow of the Balance API. It injects X-Correlation-ID if absent, so tracing starts at the edge. It adds X-Channel from the calling application's name and removes the Authorization header before forwarding, so tokens never reach MI.

OpenAPI: hand-written OAS 3.0 per API, including path parameter patterns, request schema, the success schema, and every error response (400, 401, 404, 422, 429, 502, 503, 504) referencing one shared ErrorResponse component.

API Product: Jamii Retail Banking bundles all three, with OAuth2 and the Gold tier. One subscription gives a channel app (for example the mobile app) access to all three. Partners can still subscribe to individual APIs.

Developer Portal onboarding (consumer steps)

1. Sign in at <https://portal.dev.jamii.daqiqtech.co.ke/devportal> and browse or search the jamii tag.
2. Read the API docs and try the OpenAPI console.
3. Create an application (for example jamii-mobile) and choose a business plan.
4. Subscribe the application to the Jamii Retail Banking product, or to individual APIs.
5. Under Production Keys, generate client credentials (OAuth2) or an API key.
6. Call the gateway at <https://api.dev.jamii.daqiqtech.co.ke/jamii/...> with Authorization: Bearer <token> or the apikey header.

Until the nginx stretch is delivered, the same steps run locally against <https://localhost:9443/devportal> and the gateway at <https://localhost:8243>.

8. Deployment architecture

One docker compose up -d starts the whole dev environment on a single bridge network, jamii-net. Only APIM's gateway and portal ports are meant for consumers; MI is reachable directly only for debugging.

Service	Image	Built from	Baked into the image	Host ports	Compose profile
h2	jamii/h2:\${TAG}	eclipse-temurin:21-jre + H2 2.2.224 jar	schema.sql + seed.sql, run by RunScript before the TCP server starts; database recreated on every start (H2_RESET_ON_START)	9092, 8082 (console, dev only)	core
mi	jamii/mi:\${TAG}	eclipse-temurin:17-jre + wso2mi-4.3.0.zip	Integration artifacts (hot-deployed for local runs; a CAR in the pipeline), H2 2.2.224 driver, 1 s timeout handler, log-masking layout (configure.sh)	8290, 8253, 9164	core
apim	jamii/apim:\${TAG}	eclipse-temurin:17-jre + wso2am-4.3.0.zip	deployment.toml (ELK analytics on), custom policy, keystores	9443, 8243, 8280	core
filebeat	jamii/filebeat:\${TAG}	elastic/filebeat:8.x	filebeat.yml (MI and APIM inputs)	none	observability
logstash	jamii/logstash:\${TAG}	logstash:8.x	Pipeline: kv/json parse, re-mask, index routing	5044	observability
elasticsearch	stock elasticsearch:8.x	n/a	Config via env, 1 GB heap	9200	observability
kibana + kibana-setup	stock kibana:8.x + jamii/kibana-setup:\${TAG}	curl image	dashboards.ndjson imported once at start	5601	observability
jenkins	jamii/jenkins:\${TAG}	jenkins/jenkins:lts-jdk21	Maven, apictl, curl, jq, Docker CLI, plugins list	8080	ci
registry	stock registry:2	n/a	n/a	5000	ci
nginx	jamii/nginx:\${TAG}	nginx:stable-alpine	One server block per subdomain, TLS settings, IP allowlists (certificates mounted, not baked)	80, 443	core

Images carry platform configuration; API definitions do not. The three APIs and the product are imported into APIM by apictl at deploy time. That way the same jamii/apim image is reused in every environment, and only api_params.yaml changes.

The APIM and MI Dockerfiles unpack the downloaded zips from dist/, which is gitignored because the files are large. Each zip's SHA-256 is checked against dist/checksums.txt before the build. The containers run as a non-root wso2 user. Jenkins reads the same zips from a mounted dist volume, so CI builds exactly what runs locally.

`docker compose --profile core up -d` is enough for the API demo. Add `--profile observability` for ELK, and `--profile ci` for Jenkins and the registry.

8.1 Self-managed responsibilities (Option C)

Because both APIM and MI are self-managed, every operational concern below is ours. The table shows how this project handles each one, and what would change for production.

Responsibility	This project (dev)	Production
Hosting and scaling	docker-compose, one node per component	Kubernetes with WSO2 Helm charts; 2+ gateway and MI replicas
Product updates	Downloaded GA distributions, version pinned, checksum verified	WSO2 Updates (subscription) applied by rebuilding images
APIM internal databases	Built-in H2 (APIM default)	External RDBMS for apim_db and shared_db
Keystores and TLS	Default keystores replaced with project self-signed certs	CA-signed certificates, secure vault for secrets
Admin credentials	admin/admin changed via env vars and Jenkins credentials	IdP-backed users, least-privilege roles
Backups	Not needed (disposable stack)	Scheduled DB and config backups, tested restore
Monitoring	ELK dashboards	ELK plus alerting and on-call

Environments: each environment is a set of config values, not separate code.

Setting	dev	prod (conceptual)
apictl env name	dev	prod
APIM endpoints	apim.dev.jamii.daqiqtech.co.ke	apim.jamii.daqiqtech.co.ke
MI backend URL in API params	http://mi:8290	prod MI load balancer
Backends	Jamii mock backends (Spring Boot, mock-backends service)	real core/credit systems
Deployment trigger	automatic on main	manual approval

API endpoints differ per environment through `apictl api_params.yaml`, so the same API archive is promoted unchanged.

8.2 Observability (ELK), stretch

Logs are masked at the source in MI, shipped by Filebeat, parsed and re-checked by Logstash, stored in Elasticsearch, and explored in Kibana. ELK never receives an unmasked account number.

1. MI writes its masked key=value mediation logs and access logs to a shared volume. APIM, with analytics type `elk` in `deployment.toml`, writes gateway events to `apim_metrics.log`.
2. Filebeat tails both volumes and tags each event with its source (`mi` or `apim`).
3. Logstash parses MI lines with the `kv` filter and APIM events as JSON. It sets `@timestamp`, re-masks anything matching `\b\d{10,16}\b` to its last 4 digits, and routes events to `jamii-mi-*` or `jamii-apim-analytics-*`.
4. Elasticsearch runs as a single node with a 1 GB heap; an ILM policy deletes indices after 7 days.
5. `kibana-setup` imports data views and the dashboard on first start.

Dashboard panel	Source	Question it answers
Requests per API per minute	APIM analytics	Who is calling what, and how much
Error rate by error code	MI logs	What is failing, and why
p95 latency per API and backend	MI latencyMs	Which backend is slow
Throttled (429) calls by application	APIM analytics	Is a tier too tight
Trace by correlation ID (saved search)	MI + APIM	What happened to this one request

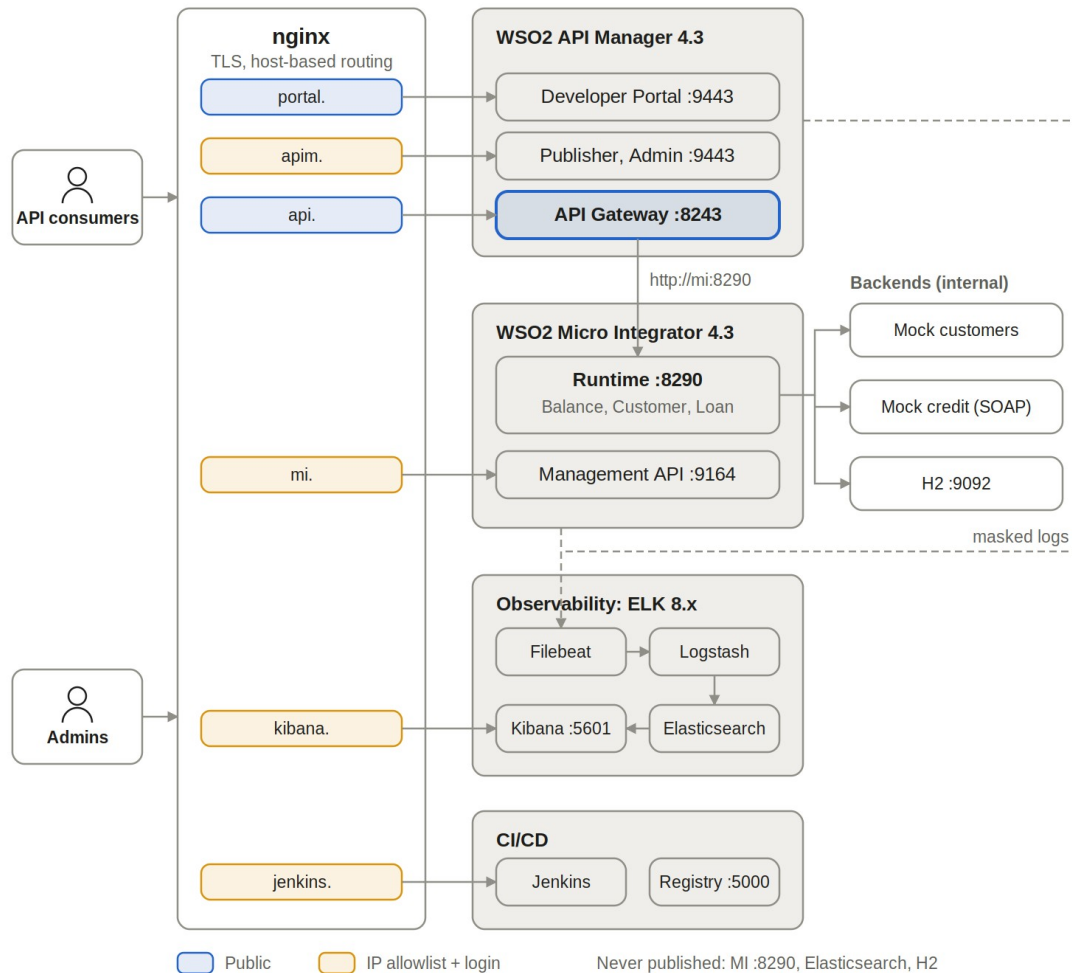
Resource budget: APIM 2 GB, Elasticsearch 1.5 GB, Logstash 1 GB (512 MB heap), MI 1 GB, Kibana 0.7 GB, Jenkins 1 GB, others under 0.5 GB. That is about 8 GB, so a 16 GB machine is the minimum. If memory is tight, Logstash can be dropped and Filebeat can write to an Elasticsearch ingest pipeline instead.

8.3 Edge topology: nginx and subdomains, stretch

nginx is the single public entry point. It terminates TLS with one wildcard certificate and routes by hostname, so no WSO2, Elastic or Jenkins port is ever exposed directly.

nginx is the only public entry, and each service gets its own subdomain

One wildcard TLS certificate: *.jamii.daqiqtech.co.ke (dev: *.dev.jamii.daqiqtech.co.ke)



Edge topology: nginx, subdomains and internal services

Subdomain (prod)	Dev equivalent	Routes to	Exposure
api.jamii.daqiqtech.co.ke	api.dev.jamii.daqiqtech.co.ke	apim:8243 (gateway, incl. /oauth2/token)	Public
portal.jamii.daqiqtech.co.ke	portal.dev.jamii.daqiqtech.co.ke	apim:9443 /devportal	Public
apim.jamii.daqiqtech.co.ke	apim.dev.jamii.daqiqtech.co.ke	apim:9443 /publisher, /admin, /carbon	IP allowlist + login (Jenkins IP allowed for apictl)
mi.jamii.daqiqtech.co.ke	mi.dev.jamii.daqiqtech.co.ke	mi:9164 management API only	IP allowlist + login
kibana.jamii.daqiqtech.co.ke	kibana.dev.jamii.daqiqtech.co.ke	kibana:5601	IP allowlist + Elastic login
jenkins.jamii.daqiqtech.co.ke	jenkins.dev.jamii.daqiqtech.co.ke	jenkins:8080	IP allowlist + Jenkins login; webhook path open

MI's runtime (8290), Elasticsearch, H2 and the registry stay on the internal Docker network. Publishing MI's runtime would let callers bypass the gateway's security and throttling.

nginx and WSO2 configuration notes

- APIM must know its public hostnames. Set the server hostname, proxy port 443, and the gateway and Developer Portal URLs in deployment.toml; otherwise logins and redirects point at internal ports. Verify the exact keys against the 4.3 docs.
- WSO2 listens on HTTPS internally with self-signed certificates, so nginx uses proxy_pass https://apim:9443 and proxy_ssl_verify off. This is acceptable on the internal network for dev; prod uses trusted certificates.
- Raise client_max_body_size (for example to 50m) on apim. so apictl API imports are not rejected.
- Set Kibana server.publicBaseUrl and the Jenkins URL to their public hostnames.
- Locally, there is no real DNS: use mkcert for the wildcard certificate and /etc/hosts entries pointing the six hostnames to 127.0.0.1.

8.4 Mock backends

The customer system and the credit-check engine are mocked by one Spring Boot service, jamii-mock-backends. It lives in mock-backends/ of the integration repository and runs as the mock-backends service of the main docker-compose.yml, so one docker compose up starts everything. It keeps its own docker-compose.yml so it can also be deployed on its own.

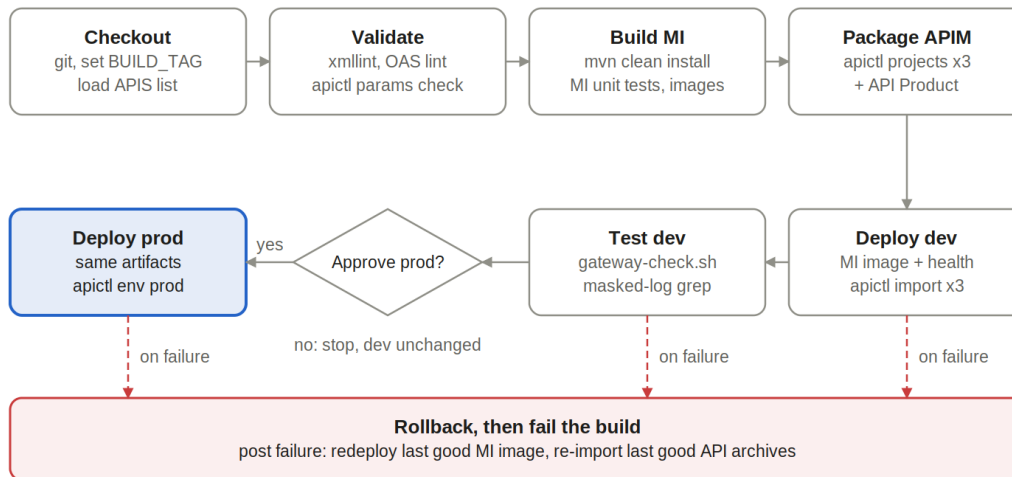
Item	Value
Stack	Java 21, Spring Boot 3.5 (Spring Web and Spring Web Services), Maven, executable jar
Endpoints	GET /customers/{id}, POST /ws (SOAP 1.1), GET /ws/creditCheck.wsdl, GET /actuator/health
API docs	Swagger UI at /swagger-ui.html and OpenAPI 3 spec at /v3/api-docs (springdoc) for the customer REST API; the SOAP contract is the WSDL, linked from the Swagger UI description
Image	jamii/mock-backends:\${TAG}; multi-stage build (maven:3.9-eclipse-temurin-21, then eclipse-temurin:21-jre), non-root user
Run	mock-backends service of the main docker-compose.yml (host port 8081 to container 8080, localhost only); MI waits for it to be healthy
Network	On the stack's own network, with the alias jamii-mock-backends, so MI calls http://jamii-mock-backends:8080
Elsewhere	Set CUSTOMER_BACKEND_URL and CREDIT_BACKEND_URL on MI to the deployed host
Logging	One line per stage, tagged with the caller's X-Correlation-ID: REQUEST_IN (method, path, client IP, headers), REQUEST_BODY, RESPONSE_OUT (status, duration, headers, body; WARN for 4xx, ERROR for 5xx), plus business lines (LOOKUP, DECISION, RULE, REJECT). Credentials, PII and amounts are masked.

No separate start step: docker compose up starts the mock backends before MI, alongside H2, MI and API Manager.

9. CI/CD pipeline design

One declarative Jenkinsfile builds and deploys all three APIs together. They are released as one unit or not at all. Every image is tagged with the build number and pushed to the local registry, so dev and prod run identical images.

Any failed stage rolls back, and prod deploys only after approval



Jenkins pipeline: stages, approval gate and rollback path

How partial deploys are prevented

1. Everything is validated and packaged before anything is deployed, so a broken spec fails the run while all environments are untouched.
2. Before deploying, the pipeline records the current MI image tag and exports the current API archives (apictl export api) as rollback artifacts.
3. Deploy dev runs the MI image swap, waits for the MI health check, then imports the three APIs in a loop with set -e. The first failure stops the loop.
4. post { failure { sh 'scripts/rollback.sh' } } restores the previous image and re-imports the saved archives, then the build is marked FAILED.
5. Prod uses the same artifacts (no rebuild), with apictl env prod and an input approval step. In this assignment it is wired as a dry run.

Credentials (APIM admin, DB password) come from the Jenkins credentials store via withCredentials and are never echoed.

10. Repository structure

```

jamii-savings-integration/
├── README.md
├── Jenkinsfile
├── docker-compose.yml          # profiles: core, observability, ci
├── docker-compose.debug.yml    # publishes MI 8290 on localhost for testing
├── .env.example                # versions, JDKs, passwords (no real secrets)
├── dist/                       # downloaded WSO2 zips + checksums.txt (zips gitignored)
├── mock-backends/              # Spring Boot mock customer REST + credit-check SOAP (JDK 21,
├── Maven, jar)
├── db/
│   ├── schema.sql
│   └── seed.sql
├── mi/                         # MI integration project (MI 4.x layout)
│   ├── src/main/wso2mi/artifacts/
│   │   ├── apis/              # BalanceAPI, CustomerAPI, LoanEligibilityAPI
│   │   ├── sequences/         # Common_InSeq, Common_HeaderHygiene, Common_FaultSeq
│   │   ├── templates/         # Common_ErrorResponse, Common_MaskedLog
│   │   ├── endpoints/         # AccountsDataServiceEP, CustomerBackendEP, CreditBackendEP
│   │   └── data-services/      # AccountsDataService.dbs
└── apim/
  
```

```

|
|   apis/
|   |   JamiiAccountBalance-v1/ # apictl project (api.yaml, OAS, policies)
|   |   JamiiCustomerProfile-v1/
|   |   JamiiLoanEligibility-v1/
|   |   products/JamiiRetailBanking/
|   |   policies/jamii-gateway-context/
|   |   throttling/LoanCheck10PerMin.json
|   |   params/ # api_params.yaml per environment
|   |   docker/ # one Dockerfile + config per image
|   |   |   h2/ # entrypoint: schema + seed, TCP server
|   |   |   mi/ # Dockerfile, entrypoint.sh, configure.sh
|   |   |   apim/ # deployment.toml (ELK analytics)
|   |   |   filebeat/
|   |   |   logstash/ # pipeline/jamii.conf
|   |   |   nginx/ # conf.d per subdomain; certs gitignored
|   |   |   kibana-setup/ # dashboards.ndjson
|   |   |   jenkins/ # plugins.txt, tools
|   |   tests/
|   |   |   gateway/ # gateway-check.sh settings per environment
|   |   |   schemas/error-response.schema.json
|   |   scripts/ # mi-smoke.sh, gateway-check.sh, build-images.sh, deploy-*.sh,
|   |   rollback.sh
|   |   docs/ # BRD, design doc, diagrams, Postman, Jira plan

```

The Jenkinsfile loops over a single list, APIS = ['balance','customer','loan']. Adding a fourth API means adding a folder and one list entry, with no new pipeline code.

11. Testing strategy

Tests run in the pipeline in order of cost: fast checks first, full gateway tests last.

Level	Tool	What it proves	Pipeline stage
Static	xmllint, OAS lint (spectral), apictl validation	Artifacts and specs are well formed	Validate
Unit	JUnit in the mock backends (17 tests, including log masking)	Masking rules work in isolation	Build
API behaviour	MI unit testing framework (test suites in each MI module, mock services for the REST and SOAP endpoints, H2 container for Balance)	Each API: happy path, 400, 404, 422/502 SOAP fault mapping, backend down	Build (mvn clean install)
Contract	JSON-path assertions in every MI error test (\$error.code, \$.error.httpStatus, \$.requestId, \$.timestamp)	Every error body has the same shape	Build
Gateway (end to end)	scripts/gateway-check.sh (curl, exit codes) via APIM	401 without token, 200 with token, 429 over tier, product subscription works	Test (dev)
MI smoke	scripts/mi-smoke.sh (19 checks, MI direct)	Every success and error path, both timeouts, correlation ID, no leaked headers, SOAP XML or stack traces, standard error body	Test (dev)
Security/logging	grep on MI logs	No full account number or PII leaked, in log files or in Elasticsearch	Test (dev)

The MI unit tests need an MI runtime in CI. The Jenkins agent unpacks the MI zip from the mounted dist/ volume and points the unit test plugin at it. If the MI version cannot assert on HTTP status codes directly, tests assert on \$.error.httpStatus in the body.

Timeout scenarios, which MI mock services do not simulate well, use the mock backends' slow rules: customer 5004 waits 7 s against a 5 s timeout, and credit customer 5004 waits 12 s against 10 s. docker stop h2 covers the database. The 503 and 504 paths are tested on purpose, not by luck.

12. Design decisions, trade-offs and bonus design

ADR	Decision	Alternative rejected	Trade-off accepted
01	H2 in server mode, in its own container	H2 embedded inside MI; MySQL	Not production-grade, but light; a separate container still lets the demo stop the DB to show 503
02	Spring Boot mock backends (customer REST and credit-check SOAP) in mock-backends/, run as part of the stack	Public mocks (JSONPlaceholder, Beeceptor, DNE Calculator)	One more service to build and run, and not an existing public service, so the README justifies it; in return, full control of data, faults and delays, with no quotas or third-party outages
03	Credit-check SOAP contract (CheckEligibility) designed for lending	DNE Calculator Divide	A contract we own must be documented (XSD and WSDL published); in return the SOAP call carries real lending semantics
04	Mock backends run locally in CI on the shared Docker network	Rely on public services	More setup, but the demo and CI cannot be broken by a third party
05	Shared sequences/templates; two-layer masking (whitelisted logging plus a log-layout rule)	Java class mediator; inline logic per API	Rule-based rather than field-aware, so short IDs rely on never being logged; in return no WSO2 Java dependencies, and MI's own internal logs are covered
06	Correlation ID at both APIM (policy) and MI	Only one layer	Small duplication; MI stays safe when called directly
07	Every component as a versioned image in a local registry	Official WSO2 images with mounted config and hot-copied CARS	Longer builds; immutable releases, and rollback becomes "run the previous tag"
08	APIM config as apictl projects in Git, imported at deploy time	APIs baked into the APIM image	One more deploy step; one APIM image serves every environment
09	One API Product plus individual APIs	Product only	Two subscription paths to explain; fits channels and partners
10	Validate everything before deploying anything, then roll back on failure	Deploy each API as it builds	Longer pipeline; never a partial release
11	ELK with Logstash, masking at source plus re-masking in Logstash	No central logging; Filebeat straight to Elasticsearch	About 3 GB more RAM; searchable traces and a second masking safeguard
12	Elastic security on (password from .env), TLS off locally	Security disabled	Slightly more setup; mirrors how a bank would run it, while TLS is noted as a prod item
13	Self-managed WSO2 (Option C), images built from downloaded distributions	WSO2 managed cloud; official Docker images	We own patching and operations; full control of versions, config and image contents
14	nginx edge with one subdomain per service under jamii.daqiqtech.co.ke	Publish service ports directly; path-based routing on one host	One more container and certificate setup; a single TLS point, IP allowlists, no internal port exposed

ADR	Decision	Alternative rejected	Trade-off accepted
15	WSO2 MI and APIM 4.3.0 on JDK 17	4.4.0 on JDK 21	Not the newest release; the WSO2 start scripts accept JDK 11 to 17 only, and MI and APIM stay on the same release line
16	Artifacts hot-deployed into MI for local runs; CAR built in the pipeline	CAR only from day one	Two packaging paths to keep in step; MI ran end to end a day earlier
17	One MI integration project (MI 4.x layout)	Five Maven modules	Less modular; matches the MI tooling layout, so it opens directly in the VS Code extension
18	Synapse timeout handler runs every 1 s	Default 15 s	Slightly more background work; 5 s and 10 s timeouts are enforced on time (found when a 12 s backend reply beat the 10 s timeout)

Bonus A: Customer dashboard aggregation (GET /customers/{customerId}/dashboard)

- MI Scatter-Gather (or Clone + Aggregate) calls Customer and Balance in parallel, with each branch timing out at 5 s.
- Decision: **return partial data**. The response is 200 with a sections object, and a failed section is marked {"status":"UNAVAILABLE","error":{"...}}. If *both* fail, the API returns 503.
- Reason: a dashboard is read-only and still useful without one panel, so failing the whole page for one slow backend is a worse customer experience.

Bonus B: Loan application events (RabbitMQ)

- An MI RabbitMQ inbound endpoint consumes loan.applications. It validates and transforms each message, then calls /loans/eligibility and publishes the result to loan.decisions.
- Malformed or unprocessable messages are rejected without requeue and go to loan.applications.dlq through a dead-letter exchange. Transient backend errors are retried 3 times with backoff before going to the DLQ.
- How this differs from the sync APIs: there is no caller waiting, so errors cannot be returned. Instead they must be retried, parked and replayed. Idempotency (an applicationId key) also matters, because a message may be delivered more than once.